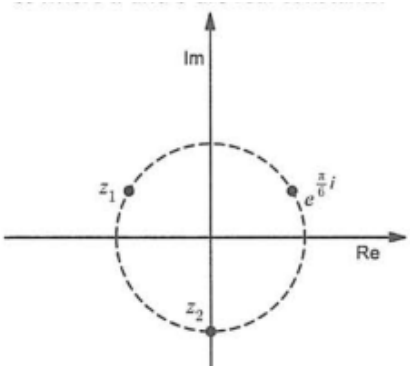
Complex Numbers: The Geometry of nth Roots

Questions 1–36

Question 1. For each given equation, sketch the solutions on an Argand diagram. In each case include a dashed line to represent the circle on which the roots lie.

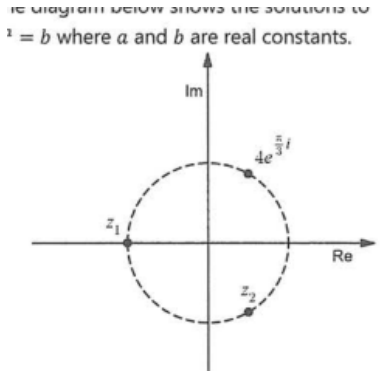
a) $z^3 = 1$
b) $z^3 = \mathrm{i}$
c) $z^5 = -\mathrm{i}$
d) $z^3 = -8$
e) $z^4 = -16\mathrm{i}$

Question 2. The diagram below shows the solutions to $z^a = bi$ where a and b are real constants.



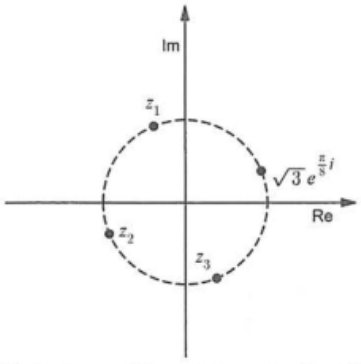
a) Write down all the solutions to $z^a = bi$ in the form $p + qi$ where p and q are real values to be found.
b) State the exact values of a and b.

Question 3. The diagram below shows the solutions to $z^a = b$ where a and b are real constants.



a) Write down all the solutions to $z^a = b$ in the form $p + qi$ where p and q are real values to be found.
b) State the exact values of a and b.

Question 4. The diagram below shows the solutions to $z^a = bi$ where a and b are real constants.



a) Write down all the solutions to $z^a = bi$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.
b) State the exact values of a and b.

Question 5. It is given that $z^3 = 8$.

a) Find the three roots of $z^3 = 8$ in the form $a + bi$ where a and b are real constants.

b) Plot the roots of $z^3 = 8$ on an Argand diagram, include a dashed line to represent the circle on which the roots lie.

c) Hence, state the three roots of the equation $(z - i)^3 = 8$ in the form $a + bi$ where a and b are real constants.

d) Plot the roots of $(z - i)^3 = 8$ on an Argand diagram.

Question 6. It is given that $z^4 = 16i$.

a) Find the four roots of $z^4 = 16i$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Plot the roots of $z^4 = 16i$ on an Argand diagram, include a dashed line to represent the circle on which the roots lie.

c) Hence, state the four roots of $(z - 2)^4 = 16i$ in the form $k + re^{i\theta}$ where $-\pi < \theta \leq \pi$, $r > 0$ and $k > 0$.

d) Plot the roots of $(z - 2)^4 = 16i$ on an Argand diagram.

Question 7. It is given that $z^6 = -64i$.

a) Find the six roots of $z^6 = -64i$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Plot the roots of $z^6 = -64i$ on an Argand diagram, include a dashed line to represent the circle on which the roots lie.

c) Hence, state the six roots of $(z + 1 - i)^6 = -64i$ giving your answer in the form $a + bi$ where a and b are given to two decimal places.

d) Plot the roots of $(z + 1 - i)^6 = -64i$ on an Argand diagram.

Question 8. Find the coordinates of the vertices of each of the following regular polygons which all have centres at the origin, giving your answers to three significant figures:

a) Triangle with one vertex at $(1, 3)$

b) Pentagon with one vertex at $(3, 2)$

c) Hexagon with one vertex at $(2, 2)$

Question 9. Find the area each of the following regular polygons which all have centres at the origin giving your answers in exact form:

a) Triangle with one vertex at $(2, 2)$
b) Square with one vertex at $(5, 1)$
c) Hexagon with one vertex at $(3, 4)$

Question 10. Find the coordinates of the vertices of a regular hexagon with a vertex at $(2, 2)$ and with centre at $(0, 3)$ giving your answers to two decimal places.

Question 11. Find the coordinates of the vertices of an equilateral triangle with a vertex at $(2, -1)$ and with centre at $(1, 4)$.

Question 12. An equilateral triangle is such that it is centred at the origin and has area $3\sqrt{3}$. One of the vertices lies on the x-axis. Determine the coordinates of the vertices of the triangle giving your answers in exact form.

Question 13. An equilateral triangle is such that it is centred at $(1, 1)$ and has area $12\sqrt{3}$. One of the vertices is directly above the centre. Determine the coordinates of the other vertices of the triangle giving your answers in exact form.

Question 14. An equilateral triangle ABC has centre at the origin and a vertex at $A(3, 4)$.

a) Find the coordinates of points B and C giving your answers to two decimal places.

b) Find the exact area of the triangle ABC.

A triangle is formed using the coordinates of the midpoints of AB, BC and CA.

c) Find the area of the triangle formed by these three midpoints giving your answer in exact form.

Question 15. It is given that $f(z) = z^3 + a$ where a, b and c are real constants. The roots of the equation $f(z) = 0$ are z_1, z_2 and z_3. When plotted on an Argand diagram, the points representing these roots form the vertices of a triangle. The area of the triangle is $27\sqrt{3}$ and $\arg(z_1) = \frac{\pi}{3}$. Determine the value of a.

Question 16. It is given that $f(z) = z^3 + az^2 + bz + c$ where a, b and c are complex constants. The roots of the equation $f(z) = 0$ are z_1, z_2 and z_3. When plotted on an Argand diagram, the points representing these roots form the vertices of a triangle centred around $1 + i$. It is known that the area of the triangle is $3\sqrt{3}$ and $\text{Im}(z_1) = 1$ and $\text{Re}(z_1) > 0$. Determine the values of a, b and c.

Question 17. It is given that $f(z) = z^3 + az^2 + bz + c$ where a, b and c are real integers. The roots of the equation $f(z) = 0$ are z_1, z_2 and z_3. When plotted on an Argand diagram, the points representing these roots form the vertices of a triangle centred around $3 + 0i$. It is known that the area of the triangle is $\frac{27}{4}\sqrt{3}$ and $\text{Im}(z_1) = 0$ and $\text{Re}(z_1) > 0$. Determine the values of a, b and c.

Question 18. It is given that $f(z) = (z - 1)^3$.

a) Find the solution to $f(z) = 27i$ giving your answers in the form $a + bi$.

It is given further that the solutions to $f(z) = 27i$ represent the points A, B and C on an Argand diagram, where A and B have positive imaginary parts and A having the smaller real part.

b) Show that the area of the quadrilateral $OABC$ where O is the origin is $\frac{9}{4} + \frac{9\sqrt{3}}{2}$.

Question 19.

a) Find the cube roots of $-5 + 7\sqrt{5}i$ giving your answer in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Plot the cube roots on an Argand diagram.

c) State the angles that any lines of symmetry of the triangle formed by the cube roots in the Argand diagram make with the positive real axis.

Question 20. Find the least value of $\arg z$ in the interval $-\pi < \arg z \leq \pi$ when $z^5 = \frac{\sqrt{3}}{2}(1 + i)$.

Question 21. Find the least value of $\arg z$ in the interval $-\pi < \arg z \leq \pi$ when $z^6 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$.

Question 22. Given two roots of $z^6 = i$, z_1 and z_2, lie in the positive quadrant of the Argand diagram. For $w = \frac{1}{2}(z_1 + z_2)$:

a) Find $\arg(w)$.

b) Hence, find the smallest value of n such that w^n is real.

Question 23. Given two roots of $z^5 = \sqrt{3} + i$, z_1 and z_2, lie in the positive quadrant of the Argand diagram. For $w = \frac{1}{2}(z_1 + z_2)$:

a) Find $\arg(w)$.

b) Hence, find the smallest value of n such that w^n is real.

Question 24. It is given that $z^3 = \sqrt{6} + \sqrt{2}i$.

a) Find the three roots of $z^3 = \sqrt{6} + \sqrt{2}i$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Plot the roots of $z^3 = \sqrt{6} + \sqrt{2}i$ on an Argand diagram, include a dashed line to represent the circle on which the roots lie.

The transformation represented by the matrix $M = \begin{pmatrix} 6 & 2 \\ 2 & 4 \end{pmatrix}$ acts on the points plotted on the Argand diagram.

c) Find the exact area of the shape formed by the transformed points.

Question 25. It is given that $z^3 = 12\sqrt{3} + 36i$.

a) Find the three roots of $z^3 = 12\sqrt{3} + 36i$ in the form $re^{i\theta}$ where $-\pi < \theta \leq \pi$ and $r > 0$.

b) Plot the roots of $z^3 = 12\sqrt{3} + 36i$ on an Argand diagram, include a dashed line to represent the circle on which the roots lie.

The transformation represented by the matrix $M = \begin{pmatrix} 6 & 2 \\ -2 & -2 \end{pmatrix}$ acts on the points plotted on the Argand diagram.

c) Find the exact area of the shape formed by the transformed points.

Question 26.

a) Find all the roots of the equation $z^6 = 27$ in the form $a + bi$ where a and b are real constants.

b) Find the area of the shape formed on the Argand diagram of the roots of $z^6 = 27$.

The transformation represented by the matrix $M = \begin{pmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{pmatrix}$ acts on the points plotted on the Argand diagram.

These new points are the solutions to the equation $f(w) = 0$ given where $f(w) = w^6 + k$ with k a real constant.

c) Find the value of k.

d) Find the area of the shape formed by the roots of $f(z) = 0$.

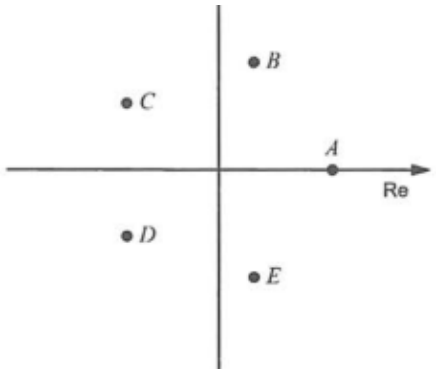
Question 27.

a) Show on an Argand diagram the points representing the three cube roots of unity.
b) Explain why, geometrically, the sum of the three cube roots of unity is zero.
c) Find the exact roots of the equation $z^3 - 1 = i$ in the form $r(\cos\theta + i\sin\theta)$ where $-\pi < \theta \leq \pi$ and $r > 0$.
d) Show that $\sin 15° + \sin 135° = \sin 105°$.

Question 28.

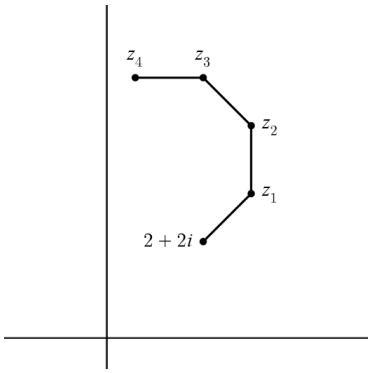
a) Find the exact roots of the equation $z^4 = 2\sqrt{3} + 2i$ in the form $r(\cos\theta + i\sin\theta)$ where $-\pi < \theta \leq \pi$ and $r > 0$.
b) Hence, show that $\cos 7.5° + \cos 82.5° + \cos 97.5° + \cos 172.5° = 0$.

Question 29. The roots of an equation are labelled A, B, C, D and E, and are shown in the diagram below. The distance between the origin and any of the points is 3 units.



a) State the equation of which the five roots are the solution.
b) Find the five roots of the equation giving your answers in the form $z = r(\cos\theta + i\sin\theta)$ for $-\pi < \theta \leq \pi$.
c) Show that a quadratic equation satisfied by B and E is $z^2 - 6z\cos\frac{2\pi}{5} + 9 = 0$.
d) Find a similar quadratic satisfied by the roots indicated by C and D.

Question 30. A particle is moving in the complex plane, with starting position $2 + 2i$. It initially faces the positive real direction. The particle rotates $\frac{\pi}{4}$ anticlockwise before moving in that direction a distance of $\sqrt{2}$. It repeats this a further three times.



Show that the displacement of the particle from its initial position is $2 - \sqrt{2} + (4 + \sqrt{2})i$.

Question 31.

a) Find the two roots of the equation $(z-1)^2 = 8 - 8\sqrt{3}i$ in the form $a+bi$ where a and b are real constants to be found.
b) Find $|z-1|^2$ for each root found in part a).
c) Find the sum of the squared distances from 1 for both roots found in part a).
d) Find the sum of the squared distances from the origin for both roots found in part a).

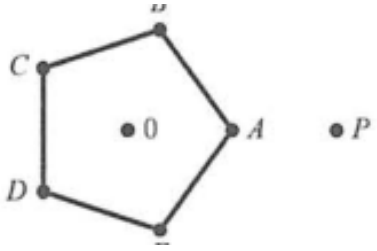
Question 32.

a) Find the roots of the equation $(z-3)^2 = -40 + 40\sqrt{3}i$ in the form $a+bi$ where a and b are real constants to be found.
b) Find $|z-3|^2$ for each root found in part a).
c) Find the sum of the squared distances from 3 for both roots found in part a).
d) Find the sum of the squared distances from the origin for both roots found in part a).

Question 33. Using the identity $zz^* = |z|^2$ show that if ω is any complex nth root of unity then $|\omega - 2|^2 = 5 - 2(\omega + \omega^*)$.

Question 34.

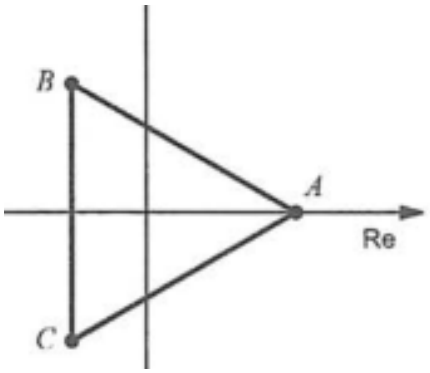
a) Using the identity $zz^* = |z|^2$ show that if ω is any complex nth root of unity then $|\omega - 3|^2 = 10 - 3(\omega + \omega^*)$.



Let $ABCDE$ be a regular pentagon whose vertices all lie on the unit circle with centre O and A at $(1, 0)$. Let the point P be the point $(3, 0)$. The squares of the distances from P to A, P to B etc. are denoted $(PA)^2 + (PB)^2 + (PC)^2 + (PD)^2 + (PE)^2$.

b) Find the value of $(PA)^2 + (PB)^2 + (PC)^2 + (PD)^2 + (PE)^2$.

Question 35. The Argand diagram below shows an equilateral triangle ABC centred about the origin where points A, B and C are complex numbers which are roots of $z^3 - a^3 = 0$.

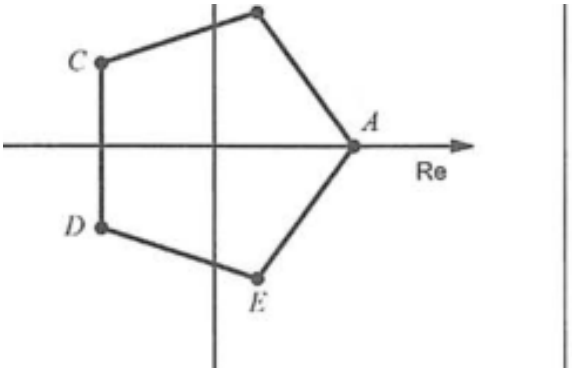


a) Find the points A, B and C in terms of a and in the form $re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$.
b) Show that the sum of the roots is zero.

A triangle is formed using the midpoints of AB, BC and CA.

c) Find a polynomial equation with roots of these midpoints. Give your equation in terms of a with integer coefficients.

Question 36. A regular pentagon $ABCDE$ centred about the origin where points A, B, C, D and E are complex numbers which are roots of $z^5 = a^5$.



a) Find the points A, B, C, D and E in terms of a and in the form $re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$.
b) Show that the sum of the roots is zero.
c) Show that $e^{i\alpha} + e^{i\beta} = 2\cos\left(\frac{\alpha-\beta}{2}\right) e^{i(\alpha+\beta)/2}$.

A pentagon is formed using the midpoints of BC, CD, DE, EA and AB.

d) Show that a polynomial equation with roots of these midpoints is given by $z^5 - a^5\cos^5\left(\frac{\pi}{5}\right) = 0$.

A regular hexagon $ABCDEF$ is centred about the origin and points A, B, C, D, E and F are complex numbers which are roots of $z^6 = a^6$.

